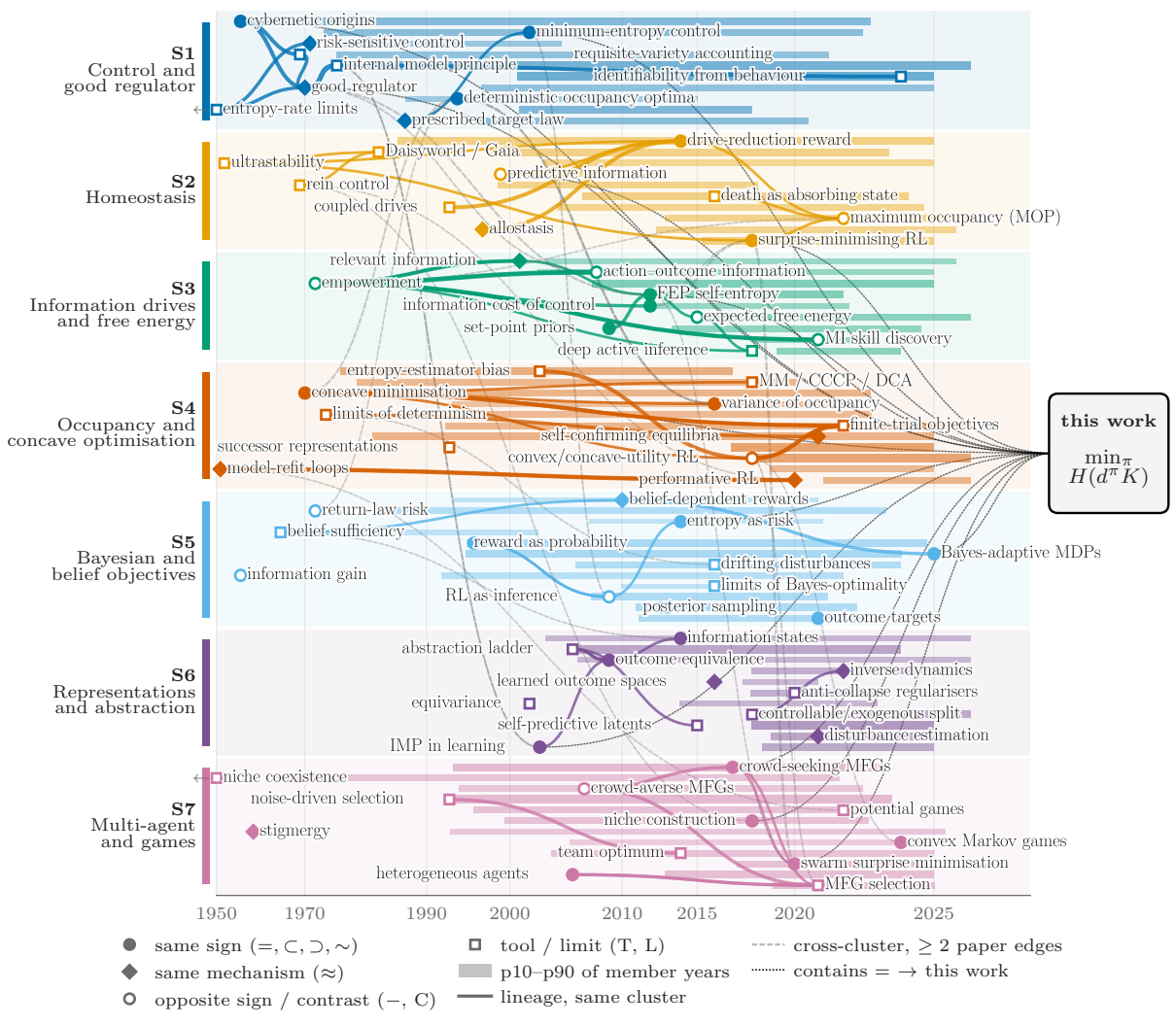

S1
Control and good regulator
cybernetic origins
risk-sensitive control
internal model principle
good regulator
entropy-rate limits
minimum-entropy control
requisite variety accounting
identifiability from behaviour
deterministic occupancy optima
prescribed target law
S2
Homeostasis
ultrastability
Daisyworld / Gaia
rein control
coupled drives
allostasis
predictive information
drive-reduction reward
death as absorbing state
maximum occupancy (MOP)
surprise-minimising RL
S3
Information drives and free energy
relevant information
empowerment
information cost of control
set-point priors
action–outcome information
FEP self-entropy
expected free energy
deep active inference
MI skill discovery
S4
Occupancy and concave optimisation
entropy-estimator bias
concave minimisation
limits of determinism
successor representations
model-refit loops
MM / CCCP / DCA
variance of occupancy
self-confirming equilibria
convex/concave-utility RL
performative RL
finite-trial objectives
this work
$\min_\pi H(d^\pi K)$
S5
Bayesian and belief objectives
return-law risk
belief sufficiency
information gain
belief-dependent rewards
entropy as risk
reward as probability
RL as inference
drifting disturbances
limits of Bayes-optimality
posterior sampling
outcome targets
Bayes-adaptive MDPs
S6
Representations and abstraction
abstraction ladder
IMP in learning
outcome equivalence
learned outcome spaces
equivariance
self-predictive latents
information states
inverse dynamics
anti-collapse regularisers
controllable/exogenous split
disturbance estimation
S7
Multi-agent and games
niche coexistence
noise-driven selection
stigmergy
heterogeneous agents
crowd-averse MFGs
team optimum
crowd-seeking MFGs
niche construction
potential games
convex Markov games
swarm surprise minimisation
MFG selection
1950
1970
1990
2000
2010
2015
2020
2025
same sign (=, ⊂, ⊃, ∼)
same mechanism (≈)
opposite sign / contrast (−, C)
tool / limit (T, L)
p10–p90 of member years
lineage, same cluster
cross-cluster, ≥ 2 paper edges
contains = → this work